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In [1]: import numpy as np
        from sklearn import datasets
        from sklearn.model_selection import train_test_split
        from sklearn.metrics import accuracy_score
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In [2]: iris = datasets.load_iris()
        X = iris.data # features
        y = iris.target # labels
```

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In [3]: X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=50
        )
```

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In [4]: class KNN:
        def __init__(self, k=3):
            self.k = k

        def fit(self, X, y):
            self.X_train = X
            self.y_train = y

        def euclidean_distance(self, x1, x2):
            return np.sqrt(np.sum((x1 - x2) ** 2))

        def predict(self, X):
            predictions = [self._predict(x) for x in X]
            return np.array(predictions)

        def _predict(self, x):
            # Compute distances
            distances = [self.euclidean_distance(x, x_train) for x_train in self.X_train]

            # Get k nearest samples
            k_indices = np.argsort(distances)[:self.k]

            # Get labels of nearest samples
            k_nearest_labels = [self.y_train[i] for i in k_indices]

            # Majority vote
            most_common = np.bincount(k_nearest_labels).argmax()
            return most_common
```

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In [5]: k = 3
        model = KNN(k=k)
        model.fit(X_train, y_train)

        predictions = model.predict(X_test)
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In [6]: accuracy = accuracy_score(y_test, predictions)
        print("Accuracy:", accuracy)
```

Accuracy: 0.9666666666666667